13 According to the Office for National Statistics, in 2022 the proportion of people in England and Wales aged 16 or over who were married or in a civil partnership was 0.494.

In 2024 a researcher collected a random sample of 78 people aged 16 or over.

The researcher decided to conduct a hypothesis test at the 1% level to see if there was any evidence to suggest that the proportion of people aged 16 or over who are married or in a civil partnership was lower in 2024.

(a) State the hypotheses for the test. You must define the parameter used to carry out the test. **[2]**

(b) Determine the critical region for the test. **[3]**

The researcher found that 26 of the 78 people in the random sample were married or in a civil partnership.

(c) Use your answer to part **(b)** to carry out the hypothesis test. **[3]**

(d) State the probability that the null hypothesis is incorrectly rejected. **[1]**